

# CPU Scheduling Simulator (Round Robin vs SJF)

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## Project Description

This project is a CPU Scheduling Simulator that compares two scheduling algorithms:

- Round Robin (RR)
- Non\_Preemptive Shortest Job First (SJF)

The system allows users to input multiple processes with arrival time and burst time, then visualizes execution using Gantt charts and calculates performance metrics such as Waiting Time, Turnaround Time, and Response Time.

The main goal is to analyze fairness vs efficiency and study the effect of time quantum in Round Robin scheduling.

## Features

- Accept dynamic number of processes
- Input validation (arrival time, burst time, quantum)
- Round Robin scheduling implementation
- Shortest Job First scheduling implementation
- Gantt chart visualization for both algorithms
- Calculation of:

- Waiting Time (WT)
- Turnaround Time (TAT)
- Response Time (RT)
- Comparison between both algorithms
- Error handling for invalid input

## How to Run

1. Clone the repository: <https://github.com/Anton-M0/CPU-Scheduler-Comparison>  
git clone
2. Open the project in your IDE ( VS Code)
3. Install required dependencies (python)
4. Run the main file: python main.py

## Project Structure

src/  
model/  
scheduler/  
metrics/  
gui/  
util/  
screenshots/  
test-cases/

## Test Scenarios

- Scenario A: Basic workload

- Scenario B: Short job heavy
- Scenario C: Fairness test
- Scenario D: Long job sensitivity
- Scenario E: Validation test

## **Conclusion**

The project demonstrated the differences between Round Robin and Non-Preemptive SJF scheduling algorithms.

Round Robin provided better fairness and responsiveness because CPU time was shared equally among processes using a time quantum.

Non-Preemptive SJF achieved lower average waiting and turnaround times for short processes, making it more efficient in many workloads.

The experiment also showed that the choice of algorithm depends on system requirements such as fairness, responsiveness, and execution efficiency.

## **Screenshots**

Check the /screenshots folder for:

- GUI
- Gantt Charts
- Results Tables
- Validation Errors